

The set average of derivative generators

MathlibAnnex.Plucker.derivativeAverage

def

Separates the volume-scaled integrand from the normalization of its average.

Statement

Let $n, N \in \mathbb{N}_0$ and give $E = \mathbb{R}^n$ and $F = \mathbb{R}^N$ their coordinate sup norms. Let λ_n be the fixed coordinate Lebesgue measure on E . Let $p : E \rightarrow \mathbb{R}$ be a continuous real seminorm with specified constants $a_p, b_p > 0$ such that

$$a_p \|x\|_\infty \leq p(x) \leq b_p \|x\|_\infty \quad (x \in E).$$

The lower bound makes p a norm. Its closed unit ball and real reference volume are

$$K_p = \{x : p(x) \leq 1\}, \quad v_p = \lambda_n(K_p) \in (0, \infty).$$

Here and below a finite Lebesgue measure is read as a real number when used as a scalar.

Let $\mathcal{I}_{n,N}$ be the finite set of increasing n -tuples of distinct rows from $\{1, \dots, N\}$. The minor-coordinate space $\mathbb{R}^{\mathcal{I}_{n,N}}$ carries its sup norm. For a linear map $B : E \rightarrow F$, put $B_{ij} = (Be_j)_i$ in the standard coordinate basis. Write $\Pi_n(B)$ for its maximal-minor vector and $\mathcal{V}_p(B)$ for that vector scaled by v_p . Their coordinates and scaling are

$$\Pi_n(B)_I = \det(B_{i_a,j})_{a,j=1}^n, \quad \mathcal{V}_p(B) = v_p \Pi_n(B) \quad (I = (i_1 < \dots < i_n)).$$

For any function $f : E \rightarrow F$, write $Df(x)$ for its Fréchet derivative where it exists and for the zero linear map otherwise. Write $G_{p,f}$ for its derivative-generator function and $\mathcal{A}_p(f)$ for the average of that function over K_p . The integrand and normalization are

$$G_{p,f}(x) = v_p \Pi_n(Df(x)), \quad \mathcal{A}_p(f) = v_p^{-1} \int_{K_p} G_{p,f}(x) d\lambda_n(x).$$

The integrals use the fixed reference measure, even when distances are measured with p .

Definition

The separate derivative-generator definition fixes the entire integrand, coordinate by coordinate:

$$G_{p,f}(x)_I = v_p \det((Df(x)e_j)_{i_a})_{a,j=1}^n.$$

The average uses the probability measure $v_p^{-1} \lambda_n|_{K_p}$, so the multiplier inside the integrand and the normalization outside it are distinct. If $G_{p,f}$ is integrable, evaluating a coordinate commutes with integration and gives

$$\begin{aligned}
(\mathcal{A}_p(f))_I &= v_p^{-1} \int_{K_p} v_p \Pi_n(Df(x))_I d\lambda_n(x) \\
&= \int_{K_p} \Pi_n(Df(x))_I d\lambda_n(x).
\end{aligned}$$

Assumptions

The source and target are finite real coordinate spaces with sup norms. The seminorm p is continuous and has the stated positive lower and upper comparison constants. The measure is the original coordinate Lebesgue measure. No Lipschitz, differentiability or integrability condition on f is part of this definition. The dimensions n, N may be zero. When $n = 0$, the unique empty minor has determinant 1; when $N < n$, the minor-index set is empty and its function space has one element. Neither case is excluded.

Conclusion

The result lies in $\mathbb{R}^{\mathcal{I}_{n,N}}$. Membership in a Plücker body requires additional hypotheses.

Lean uses a total Bochner integral, equal to zero when the vector integrand is not integrable. This convention does not certify integrability. Its total derivative at a nondifferentiability point is zero; the empty determinant when $n = 0$ is still 1.

Main citations

- Exact declaration and its source — `MathlibAnnex.Plucker.derivativeAverage`
- The separate derivative-generator definition — `MathlibAnnex.Plucker.derivativeGenerator`

Lean source signature (exact)

```

/-- The set average of derivative generators over the model closed unit ball. -/
def derivativeAverage {n N : ℕ} (M : EquivalentSeminorm (Fin n → ℝ))
  (f : (Fin n → ℝ) → (Fin N → ℝ)) :
  Matrix.MaximalMinorIndex n (Fin N) → ℝ :=
  f x in M.closedUnitBall, derivativeGenerator M f x ∂volume

```

This definition takes an arbitrary function f ; it assumes neither differentiability nor integrability.

In the source	Mathematical meaning
<code>{n N : ℕ}</code>	Arbitrary nonnegative integers n, N . Braces mean that Lean may infer these parameters; they are not extra assumptions.
<code>Fin n → ℝ; Fin N → ℝ</code>	The spaces $E = \mathbb{R}^n$ and $F = \mathbb{R}^N$ with their sup norms. A vector is a list of real coordinates, indexed from 0 in Lean and from 1 in the formulas here.
<code>M : EquivalentSeminorm (Fin n → ℝ)</code>	The continuous seminorm $p = M.p$, with $a_p = M.lower > 0$, $b_p = M.upper > 0$ and $a_p \ x\ _\infty \leq p(x) \leq b_p \ x\ _\infty$ for every $x \in E$.

In the source	Mathematical meaning
<code>f : (Fin n → ℝ) → (Fin N → ℝ)</code>	A function $f : E \rightarrow F$. The plain arrow $\rightarrow$ does not assert linearity or continuity.
<code>Matrix.MaximalMinorIndex n (Fin N)</code>	The index set $\mathcal{I}_{n,N}$: choose n distinct rows among the N rows and read them in increasing order. A function of this type into $\mathbb{R}$ is a minor-coordinate vector $z = (z_I)_{I \in \mathcal{I}_{n,N}}$.
<code>M.closedUnitBall; M.closedUnitBallVolume</code>	$K_p = \{x : p(x) \leq 1\}$ and $v_p = \lambda_n(K_p) > 0$, with the finite measure read as a real number.
<code>fderiv ℝ f x</code>	The real Fréchet derivative $Df(x) : E \rightarrow F$, with the zero linear map used when f is not differentiable at x .
<code>derivativeGenerator M f x</code>	The vector $G_{p,f}(x) = v_p \Pi_n(Df(x))$; its $I = (i_1 < \dots < i_n)$ coordinate is $v_p \det ((Df(x)e_j)_{i_a})_{a,j=1}^n$.
<code>volume; ∂volume</code>	The fixed reference Lebesgue measure λ_n ; the second notation specifies the measure of integration.

In the source	Mathematical meaning
<code>f x in M.closedUnitBall, ... ∂volume</code>	The set average $v_p^{-1} \int_{K_p} (\dots) d\lambda_n$, with v_p already present inside $G_{p,f}$.
<code>derivativeAverage M f</code>	The vector $\mathcal{A}_p(f) = v_p^{-1} \int_{K_p} G_{p,f}(x) d\lambda_n(x)$, not the average of the linear maps $Df(x)$.

Lean assigns the vector integral the value zero when its integrand is not integrable. The displayed definition is not a claim that this integrability condition holds.

Exact definitions used in this reading: Derivative-generator function.

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 13–18:

```
noncomputable section

open Set MeasureTheory
open scoped BigOperators ENNReal

namespace MathlibAnnex.Plucker
```

▼ Exact content identity

Declaration: `MathlibAnnex.Plucker.derivativeAverage`

Accepted content SHA-256: `ab8c5a421ca98424ccc38a7f916ad67102db45535cf6ffb5824dea08c1af5b00`

Accepted source guide SHA-256: `20e17826a5be32466058acb4334a395320dd81c492ee3dbee69d047ce30b084e`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`